EXPERIMENT 26

NAME: B.Akshay Kumar Reddy

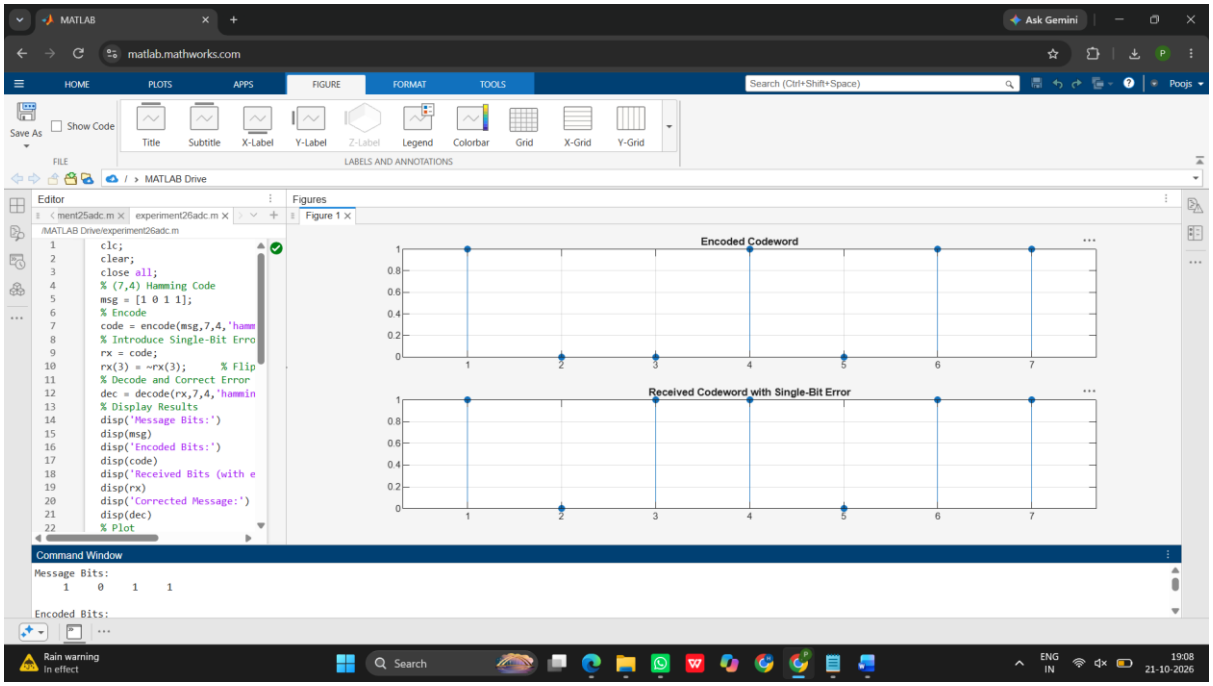
Reg no: 192412246

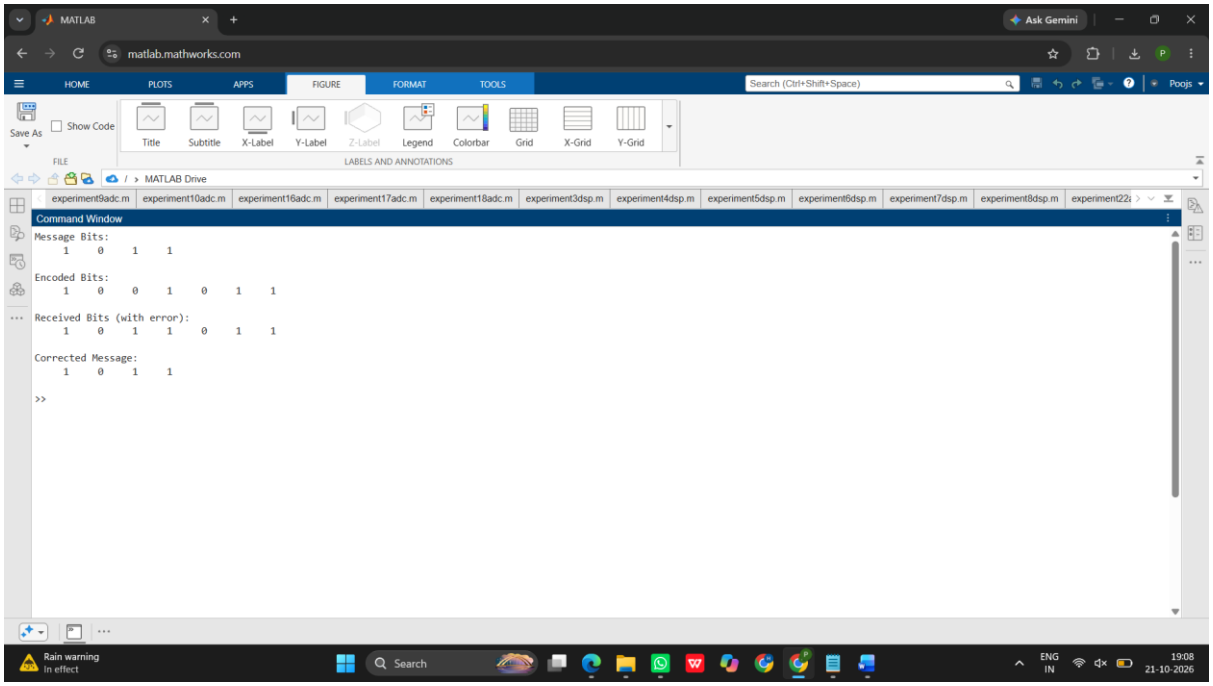
CODE:

```
clc;
clear;
close all;
% (7,4) Hamming Code
msg = [1 0 1 1];
% Encode
code = encode(msg,7,4,'hamming/binary');
% Introduce Single-Bit Error
rx = code;
rx(3) = ~rx(3);     % Flip one bit
% Decode and Correct Error
dec = decode(rx,7,4,'hamming/binary');
% Display Results
disp('Message Bits:')
disp(msg)
disp('Encoded Bits:')
disp(code)
disp('Received Bits (with error):')
disp(rx)
disp('Corrected Message:')
disp(dec)
% Plot
```

```
figure;
subplot(2,1,1)
stem(code,'filled')
title('Encoded Codeword')
grid on
subplot(2,1,2)
stem(rx,'filled')
title('Received Codeword with Single-Bit Error')
grid on
```

OUTPUT:




MATLAB
Ask Gemini
matlab.mathworks.com
HOME
PLOTS
APPS
FIGURE
FORMAT
TOOLS
Search (Ctrl+Shift+Space)
Poojs
Save As
Show Code
FILE
Title
Subtitle
X-Label
Y-Label
Z-Label
Legend
Colorbar
Grid
X-Grid
Y-Grid
LABELS AND ANNOTATIONS
MATLAB Drive
experiment9adc.m
experiment10adc.m
experiment16adc.m
experiment17adc.m
experiment18adc.m
experiment3dsp.m
experiment4dsp.m
experiment5dsp.m
experiment6dsp.m
experiment7dsp.m
experiment8dsp.m
Command Window
Message Bits:
1 0 1 1
Encoded Bits:
1 0 0 1 0 1 1
Received Bits (with error):
1 0 1 1 0 1 1
Corrected Message:
1 0 1 1
>>
Rain warning
In effect
Search
ENG
IN
19:08
21-10-2026